

# On track to discovery with the Higgs

**Abraham Tishelman-Charny**

Friday, December 12<sup>th</sup> 2025  
Stony Brook University



# HEP timeline



## LHC I

7-13.6 TeV,  $\sim 500 \text{ fb}^{-1}$

**Higgs discovery**

2009-2026



## HL-LHC II

14 TeV,  $\sim 3500 \text{ fb}^{-1}$   
**di-Higgs discovery**

2030-2041



## Future III

Up to  $\sim 100 \text{ TeV}$   
**Future discoveries**

2041 and beyond

# Next section



LHC

I

7-13.6 TeV,  $\sim 500 \text{ fb}^{-1}$

**Higgs discovery**

2009-2026



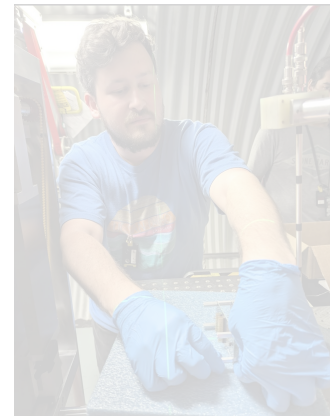
HL-LHC

II

14 TeV,  $\sim 3500 \text{ fb}^{-1}$

**di-Higgs discovery**

2030-2041



Future

III

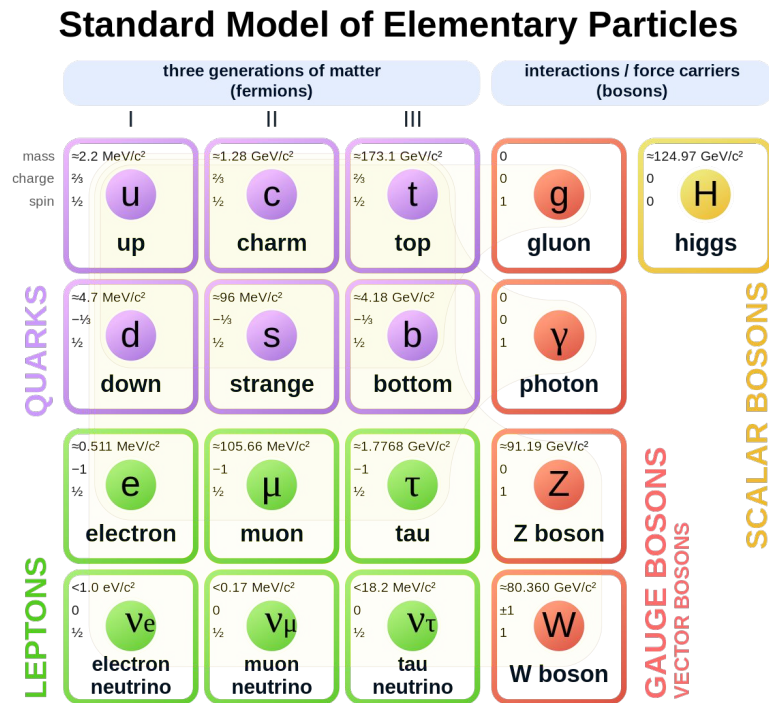
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**Future discoveries**

2041 and beyond

# The standard model

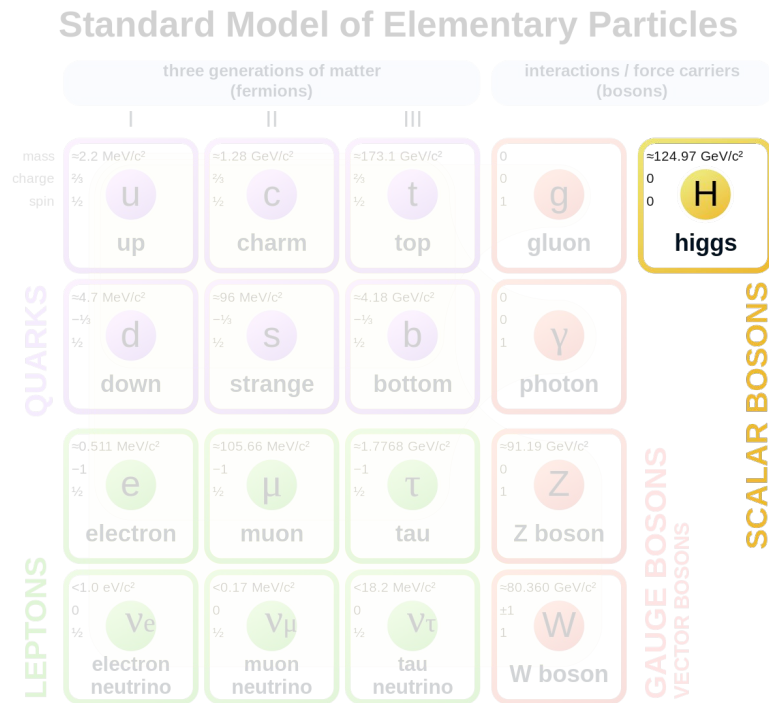
- The **standard model** (SM) of particle physics:
  - Defines **particles** and their **interactions**
- Extremely successful! Predicts **vast majority** of observed phenomena
- **Does not explain everything:**
  - Origin of **neutrino mass**
  - Nature of **dark matter**
  - **Matter-antimatter** asymmetry
  - ...





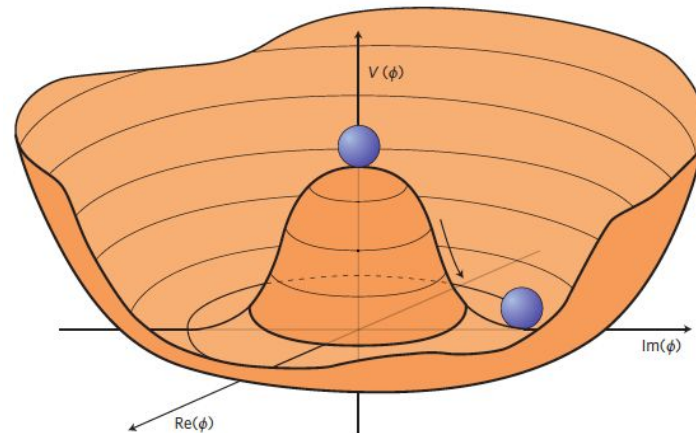
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# The Higgs boson

- **The Higgs boson:** Final SM particle discovered in 2012
- **Higgs potential:** Determines nature of Higgs interactions with **other particles**
- Intertwined with **electroweak symmetry breaking** - process by which particles acquire **mass**
- **Higgs self-coupling ( $\lambda$ )**
  - Determines magnitude of self-interaction, shape of **Higgs potential**
  - **Lacks precise measurement**
- **Has SM prediction we can compare to**



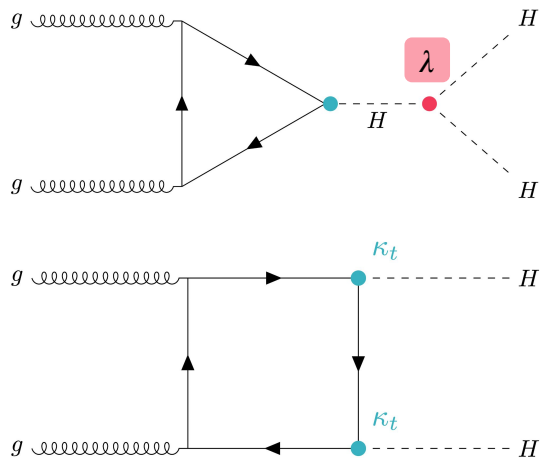
Higgs potential and mechanism

$$V(h) = V_0 + \lambda v^2 h^2 + \lambda v h^3 + \frac{1}{4} \lambda h^4 + \dots$$

# Higgs pair production (HH)

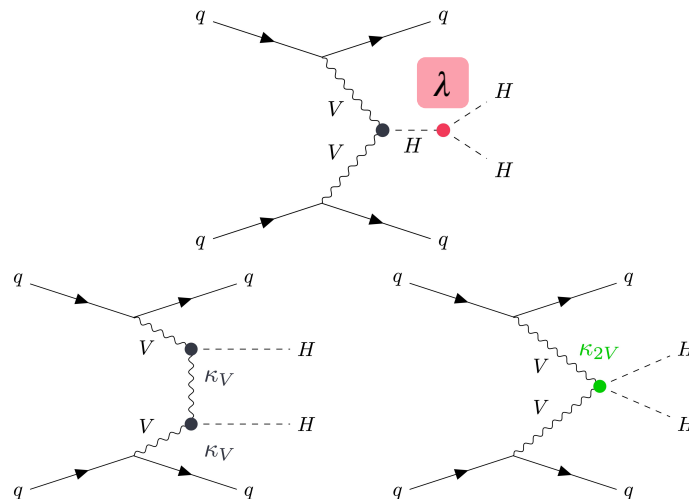
Can directly access Higgs self-coupling via **Higgs pair production (HH or di-Higgs)**:

## Gluon fusion:



$$\sigma_{\text{NNLO, FTapprox}} \sim \mathbf{31.05 \text{ fb}} @ 13 \text{ TeV}, m_H = 125.0 \text{ GeV} \quad [1803.02463]$$

## Vector boson fusion:

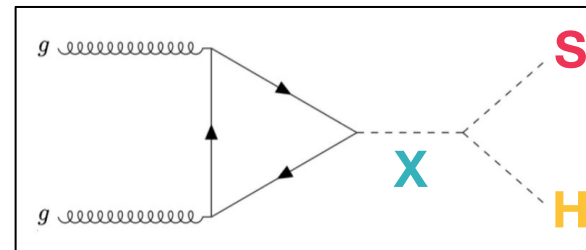
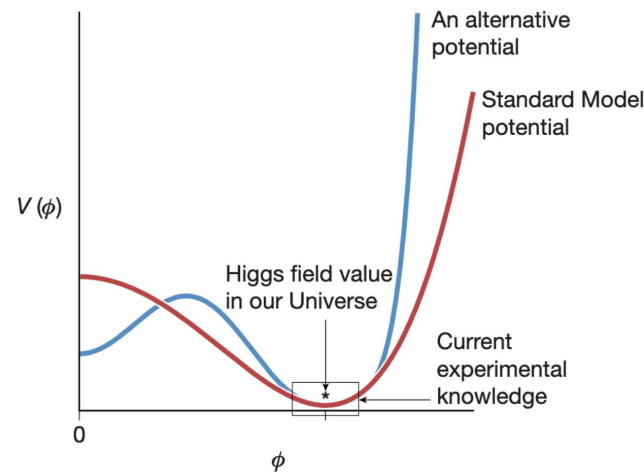


$$\sigma_{\text{N3LO QCD}} \sim \mathbf{1.73 \text{ fb}} @ 13 \text{ TeV}, m_H = 125.0 \text{ GeV} \quad [1811.07906]$$

- Self-coupling affects **production rates** and **kinematics (momentum, position)**
- About **1000 times more rare than single Higgs production**. Need **final states** with good signal to background.

# The Higgs as a portal (SH)

- Many theories predict new (BSM) particles that **interact with the Higgs boson**
- Their existence may answer our fundamental questions:
  - First order phase transition leading to **matter-antimatter asymmetry**
  - **Dark matter** candidates
  - Further testing of **Higgs potential scenarios!**
- Process considered for this coupling: “**X**→**SH**”.
  - **H**: SM Higgs
  - **X** and **S**: BSM “Higgs-like” Scalar bosons



# HH/SH final states

- **Higgs boson** has many decay modes
- Therefore, many **HH/SH** decay modes
- Most common in SM:  $H \rightarrow bb$  (and  $HH \rightarrow bbbb$ )  
(~58% at 125 GeV)
- Final states have different **likelihoods**, leave different **detector signatures**
- **SH** has two approaches:
  - **Model specific:** Model based BRs.
  - **Generic search:** Assume 100% BR.

|                | bb    | WW    | $\tau\tau$ | ZZ     | $\gamma\gamma$ |
|----------------|-------|-------|------------|--------|----------------|
| bb             | 34%   |       |            |        |                |
| WW             | 25%   | 4.6%  |            |        |                |
| $\tau\tau$     | 7.3%  | 2.7%  | 0.39%      |        |                |
| ZZ             | 3.1%  | 1.1%  | 0.33%      | 0.069% |                |
| $\gamma\gamma$ | 0.26% | 0.10% | 0.028%     | 0.012% | 0.0005%        |



= Existing SH or HH result



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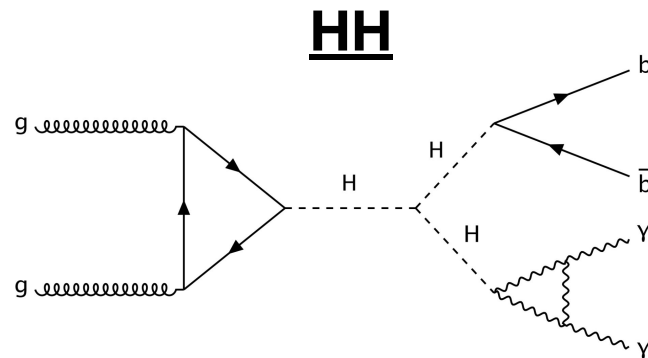
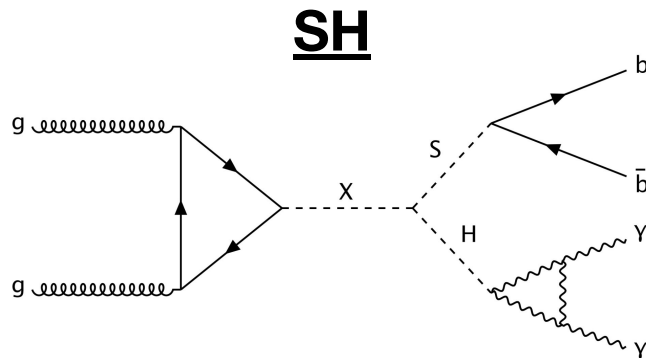
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# The $(b\bar{b})(\gamma\gamma)$ final state

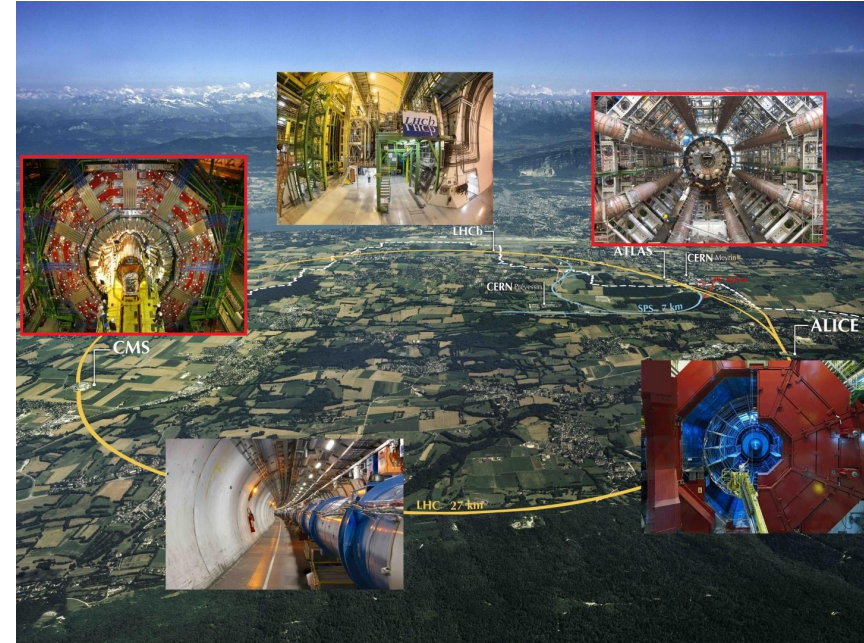
- **Clean  $\gamma\gamma$  signature.** For SM and some BSM scenarios: **High  $b\bar{b}$  branching ratio**
- HH: Can achieve good sensitivity on self-coupling measurement.
- SH: Can go very low in mass regime



- Similar signatures
- **Kinematics altered** by different **BSM** scenarios:
  - SH: Different  $X$  and  $S$  masses
  - HH: Varying of **coupling strengths**

# LHC

- Need machine capable of **producing the particles** we're looking for
- **Large Hadron Collider (LHC):** proton-proton, heavy ion collisions, pp at centre-of-mass energies of **7, 8, 13, now 13.6 TeV**



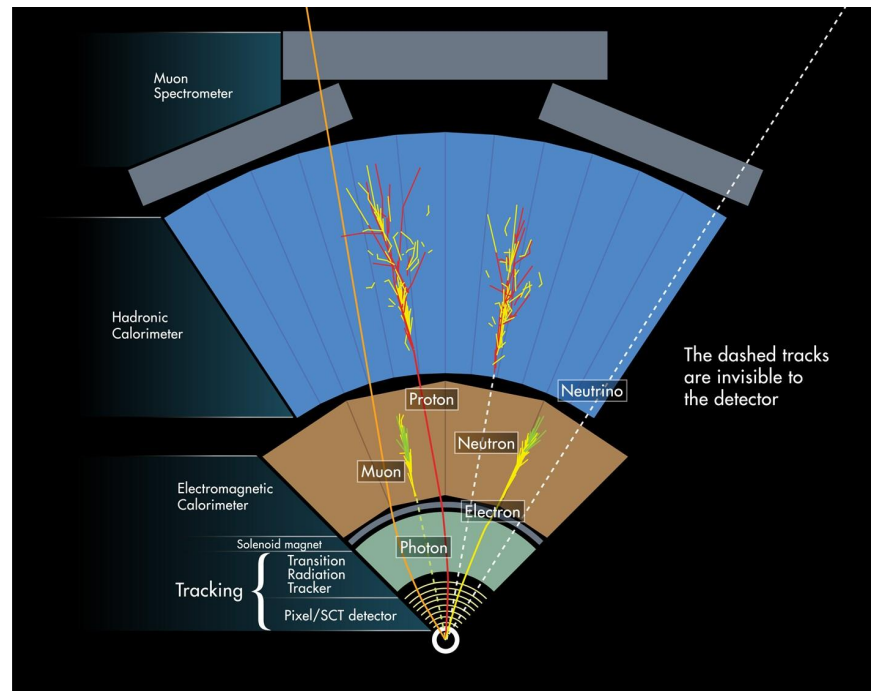
The LHC and its experiments

- Need machine capable of **producing the particles** we're looking for
- **Large Hadron Collider (LHC):** proton-proton, heavy ion collisions, pp at centre-of-mass energies of **7, 8, 13, now 13.6 TeV**
- **ATLAS & CMS:** Multipurpose detectors exploring range of physics.  
**Only detectors in the world where we can study the Higgs boson!**

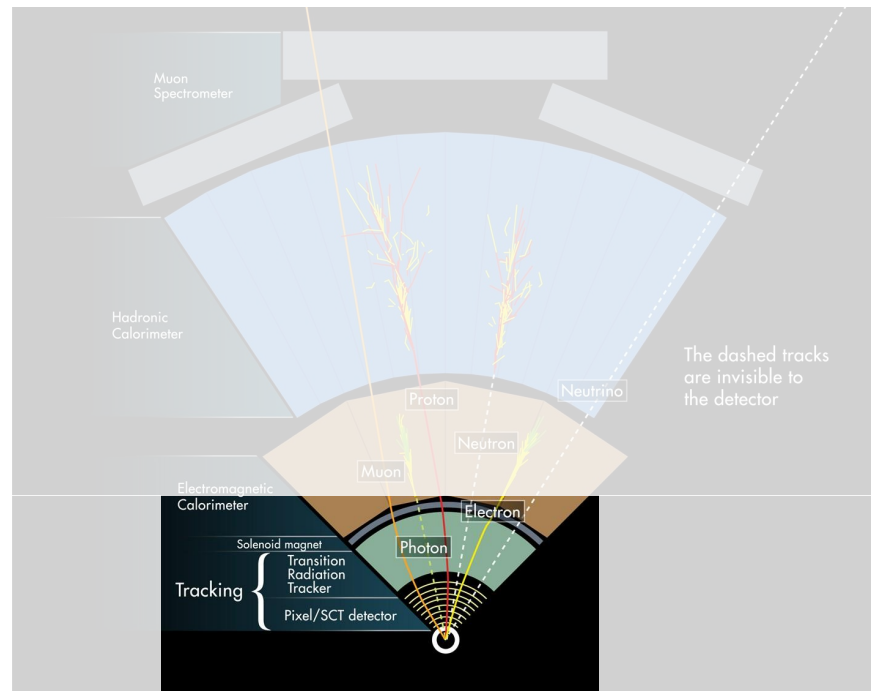


The LHC and its experiments

- Different **layers** detect different **particles** (needed for different final states!)
- Requires use of different detector **technologies** → Need to choose wisely considering **cost** and **size**
- **Reconstruct** collision by working backwards from detector information

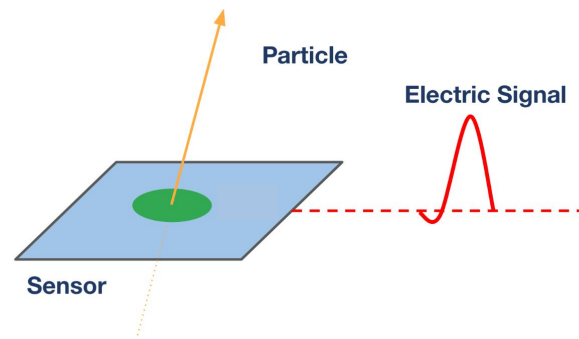


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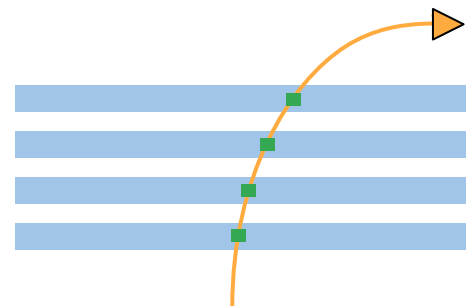


# Silicon based tracking

- Particle hits silicon
- Electric signal read out
- Series of hits forms a **track**
- Use radius to measure **particle momentum**
- Orient geometry based on magnetic field



[G. D'amen - BNL physics seminar series](#)

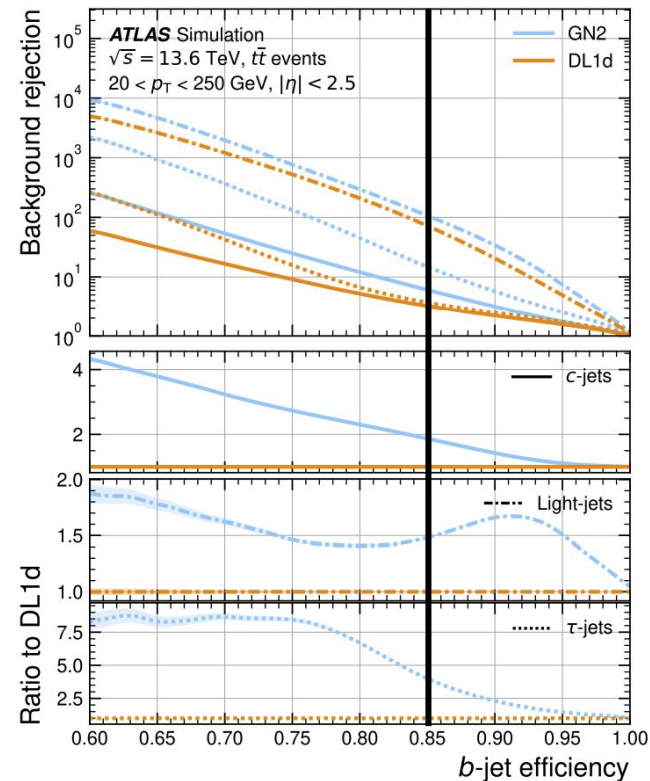


Silicon trackers actively  
used by ATLAS and CMS!

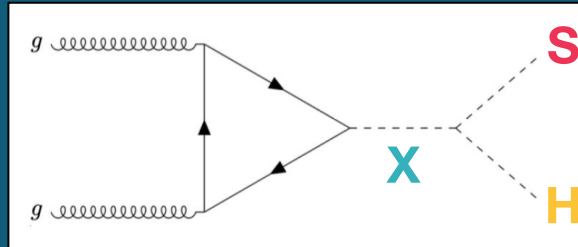


# Identifying $bb\gamma\gamma$

- SH/HH: Identify photons and b-jets with **same approaches**
- Run 2 + Run 3 analyses, **new b-tagging** in ATLAS:
  - **DL1** → **GN2**
  - **~ 8% improved b-tag efficiency for similar background rejection**
- Identify two **high momentum, isolated photons** consistent with  $H \rightarrow \gamma\gamma$ 
  - Both analyses use same **diphoton trigger**



# Run 2 + Run 3 $SH \rightarrow bb\gamma\gamma$



# Introduction

- **Signal:**

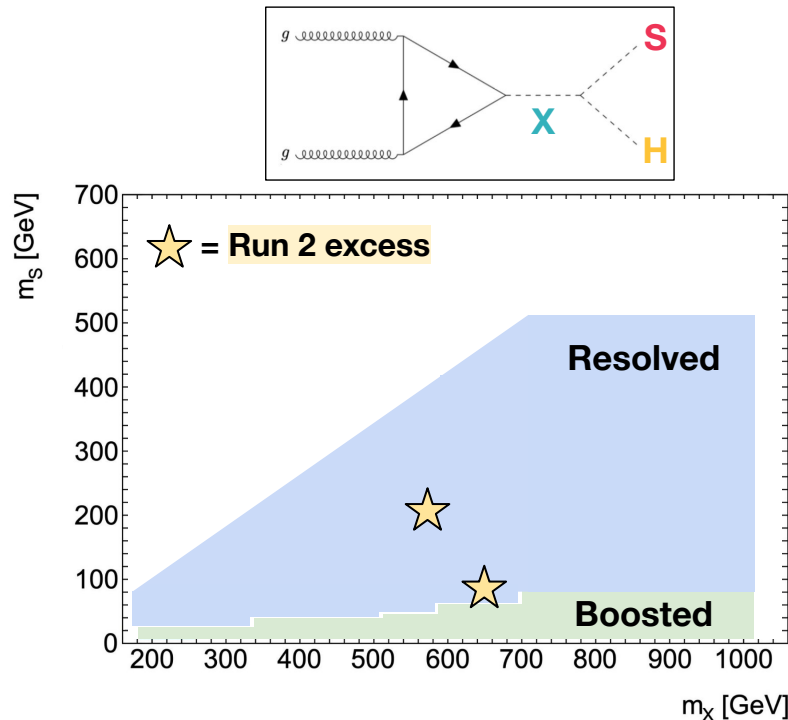
- Search **broad range** of X and S masses
- $170 < m_X < 1000$  GeV
- $15 < m_S < 500$  GeV
- Model **generic resonance** with simulation

- **Regions:**

- **Boosted** ( $m_S / m_X < 0.09$ )
- **Resolved** ( $m_S / m_X > 0.09$ )

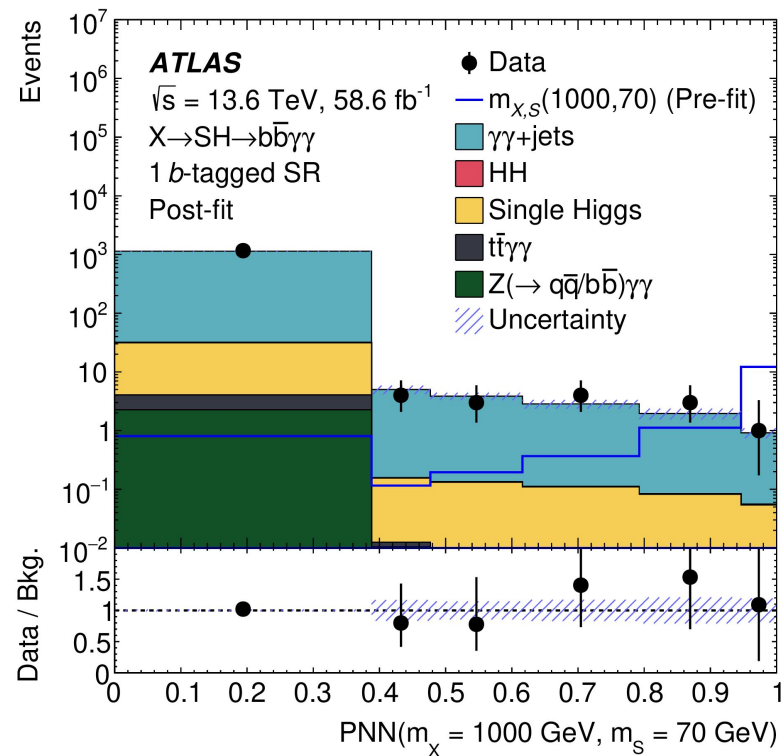
- **Excesses seen by Run 2 analyses:**

- [ATLAS](#): Local (global) **3.5 (2.0) $\sigma$**  at  $(m_X, m_S) = (575, 200)$  GeV
- [CMS](#): Local (global) **3.8 (2.8) $\sigma$**  at  $(m_X, m_S) = (650, 90)$  GeV



# PNN

- Main background process: “ $\gamma\gamma$ +jets”
- Use a **Parameterized Neural Network (PNN)** to separate signal and background.
- **360 signal mass points:** With single PNN, parameterize on X, S mass.
- Train separate PNNs for **boosted/resolved, Run 2/3**
- The PNN does a good job of separating **signal** and **background**

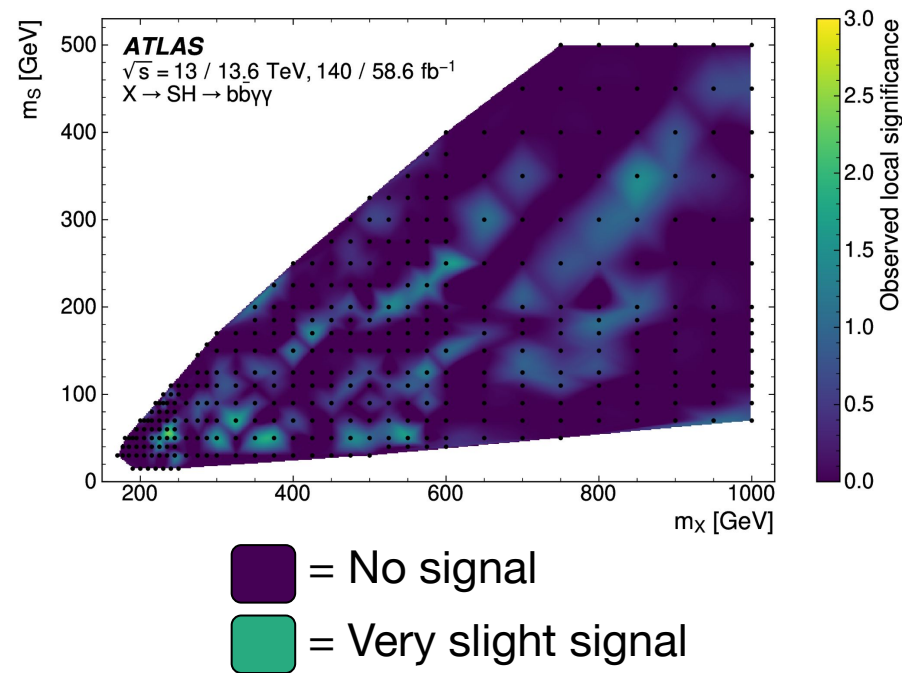


# Results

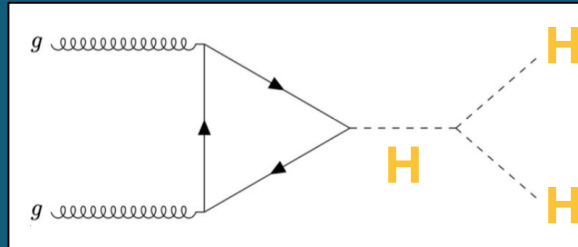
- **No significant excess observed:**
  - Maximum **local** significance:  $2\sigma$
  - Global significance  $\sim 0\sigma$  everywhere
- **Do not confirm previous Run 2 excesses from ATLAS and CMS**
- Improve sensitivity by up to **73%**
- **First Run 3 search for BSM particles coupling to the Higgs!**

[arXiv:2510.02857](https://arxiv.org/abs/2510.02857)

## Run 2 + Run 3 observed significance

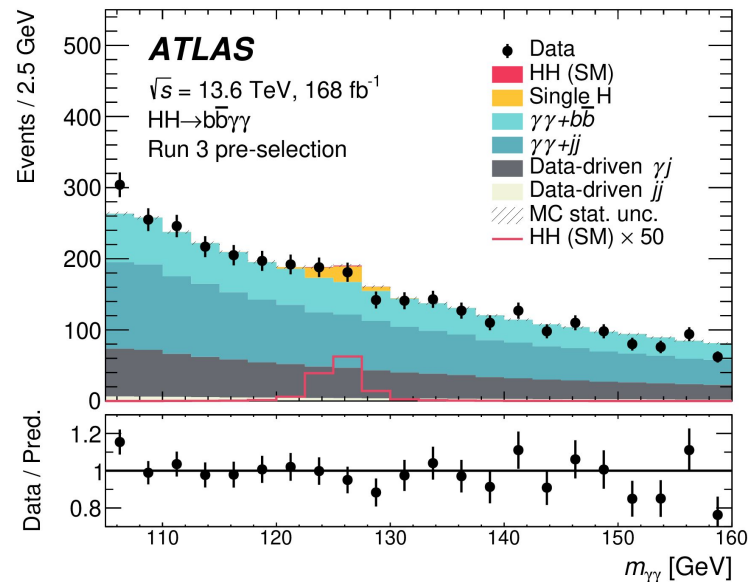


# Run 2 + Run 3 $HH \rightarrow bb\gamma\gamma$



# Signal/background modeling

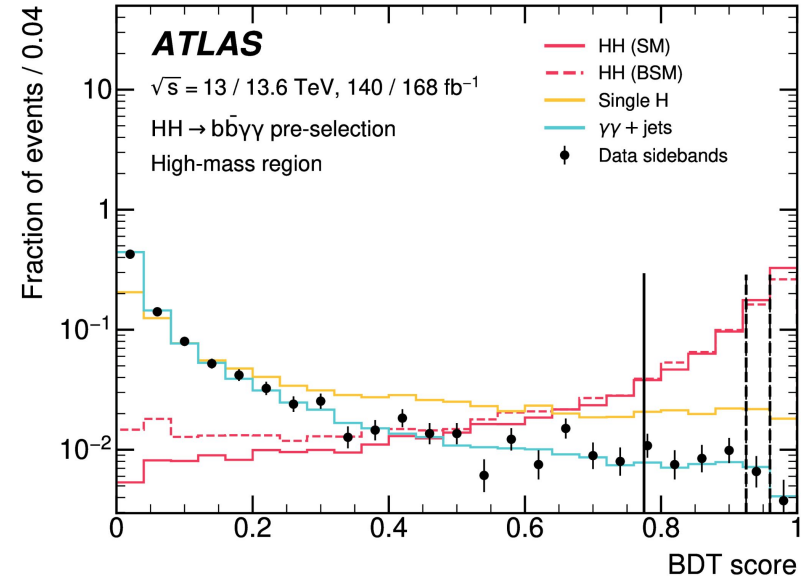
- **Signal:** HH
- **Resonant background:** Single Higgs
- **Non-resonant background:** Mainly  $\gamma\gamma$ +jets  $\rightarrow$  Same as for SH!
- Separate into Low/High mass regions (boundary at  $m_{HH} = 350$  GeV)
- Perform **kinematic fit** to further improve signal resolution





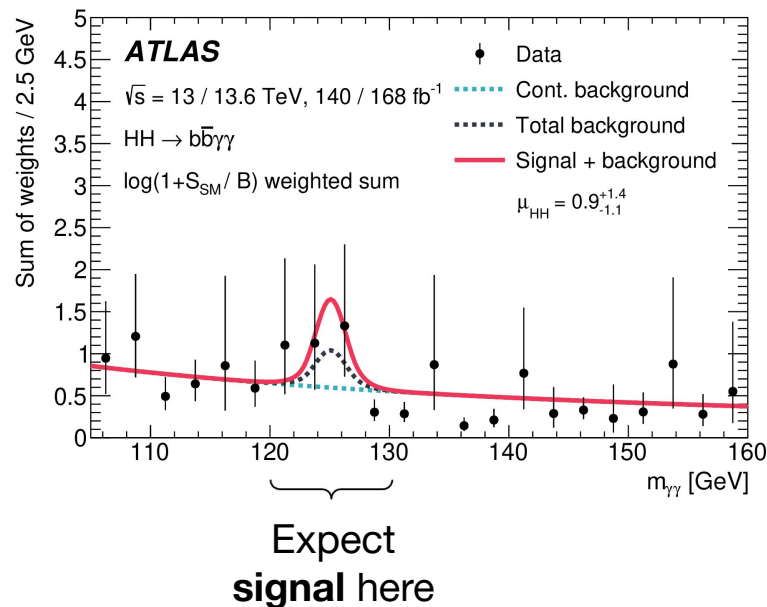
# BDT

- Use ML like in SH, but a **BDT** (no need for PNN)
- Separate BDT for Low/High mass regions:
  - **Signal:** GF and VBF HH
  - **Background:** Single Higgs and  $\gamma\gamma$ +jets
- Define 14 total categories (7 each for Run 2 and Run 3)
- Does a **good job of separating signal and background**



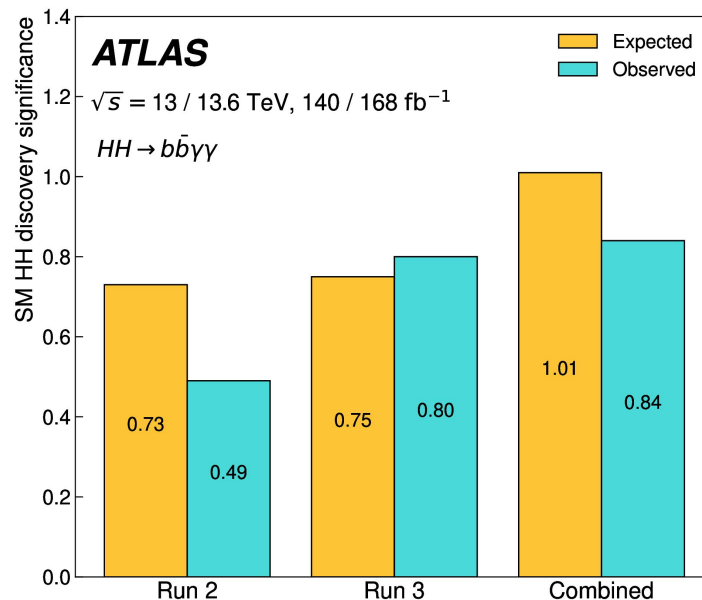
# Statistical inference

- Model non-resonant background by **fitting to data sidebands** (diphoton mass outside 120-130 GeV)
- Perform fit on **di-photon mass spectrum** - vary **HH signal strength**



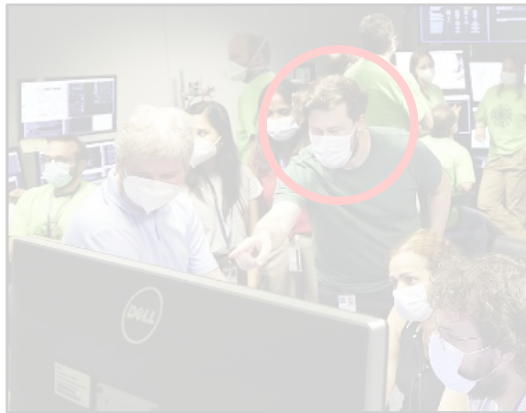
# Results

- Observed significance of  **$0.84\sigma$**  ( **$1.01\sigma$  expected**)
  - Starting to transition from **limits** to **significance!**
- Constrain self-coupling to  **$[-1.8, 6.9]$  X SM** at a 95% CL
  - Comparable to **Run 2 ATLAS+CMS combination!** [\[ATLAS-CONF-2025-012\]](#)
- First ATLAS analysis to use  $> 300 \text{ fb}^{-1}$
- First **HH analysis** to use 2024 data
- **Most sensitive single channel HH analysis to date.**



Discovery significance results

# Next section



LHC

I

7-13.6 TeV,  $\sim 500 \text{ fb}^{-1}$

**Higgs discovery**

2009-2026



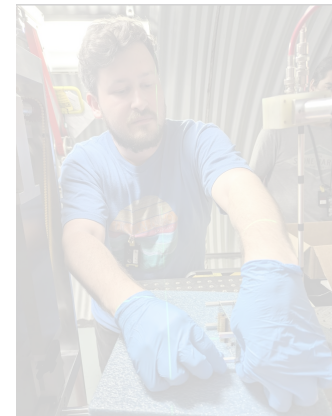
HL-LHC

II

14 TeV,  $\sim 3500 \text{ fb}^{-1}$

**di-Higgs discovery**

2030-2041



Future

III

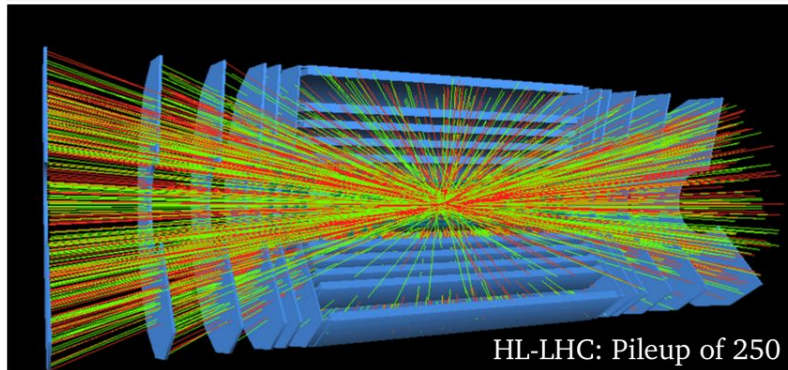
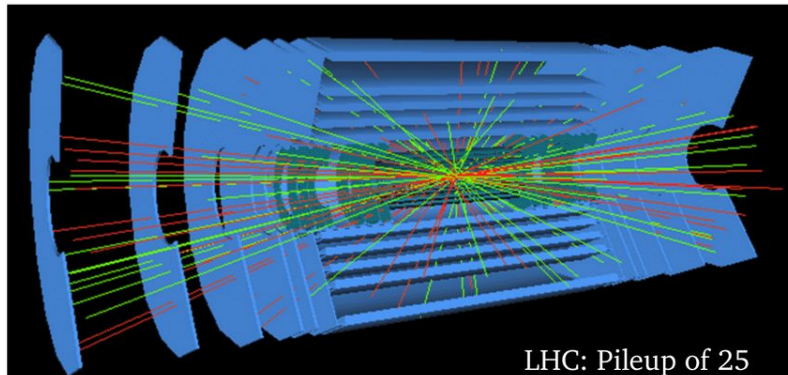
Up to  $\sim 100 \text{ TeV}$

**Future discoveries**

2041 and beyond

# HL-LHC

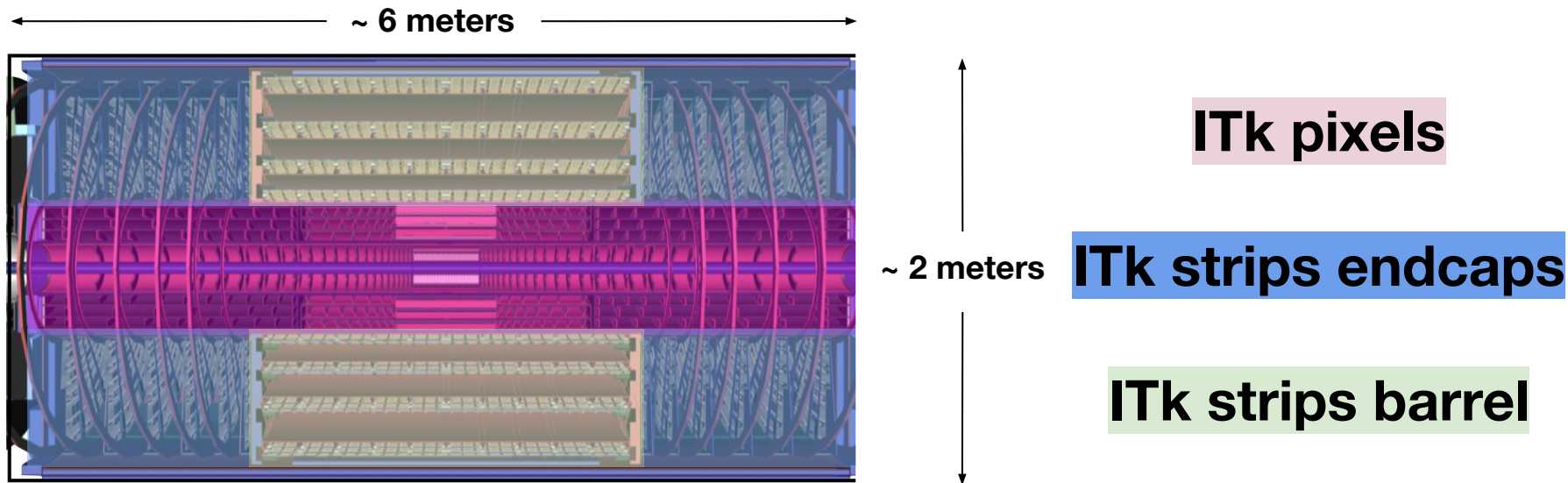
- ~2030: LHC will finish upgrade to **High-Luminosity LHC** - ready to **deliver collisions** to experiments
- **Pro:** Increase ATLAS + CMS datasets by  $\sim$  factor of 10
- **Con:** Extremely challenging data-taking environment
- **ATLAS cannot cope with this!**  
How will we enable discovery at HL-LHC?
  - **Extensive detector upgrades needed**



Development and Evaluation of Novel, Large Area,  
Radiation Hard Silicon Microstrip Sensors for the  
ATLAS ITk Experiment at the HL-LHC

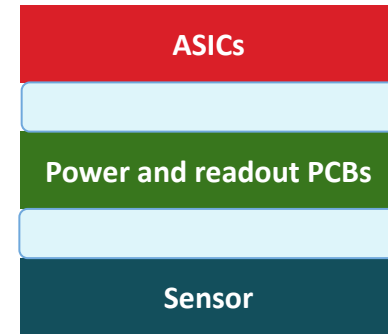
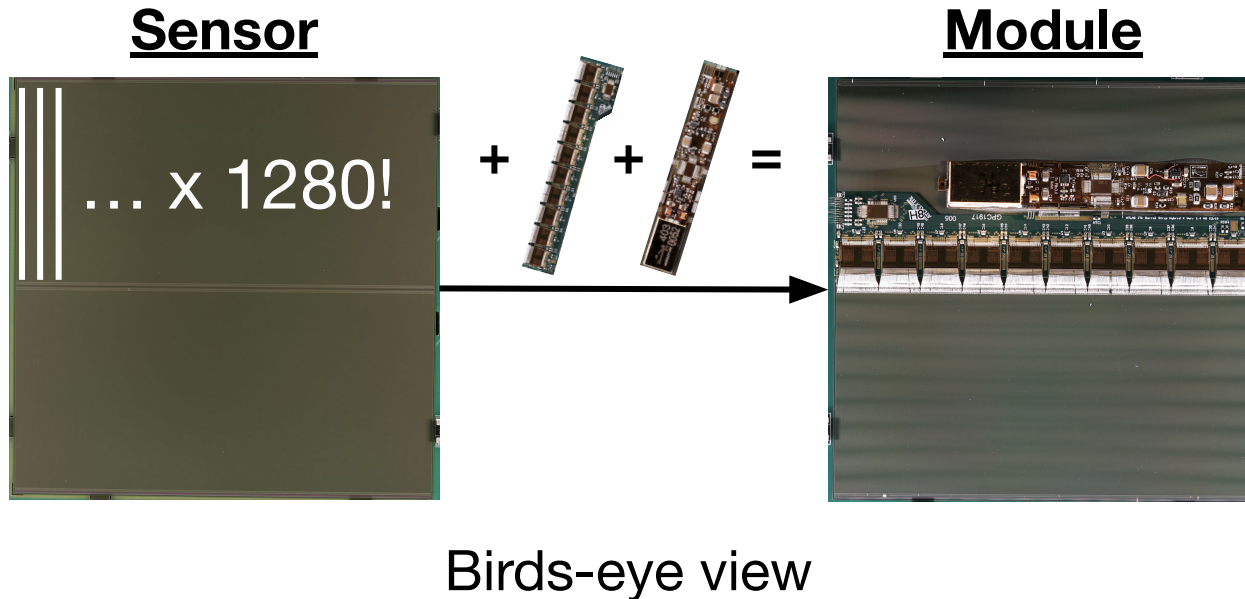
# The ATLAS ITk

- To continue excellent **b-tagging** in this environment, need a performant **tracker**
- Part of ATLAS **upgrade**: Replace tracker with **silicon pixel** and **strip** sub-detectors - **The ITk** (Inner Tracker):



# Modules

- Individual detection unit of the ITk strips: a “**Module**”

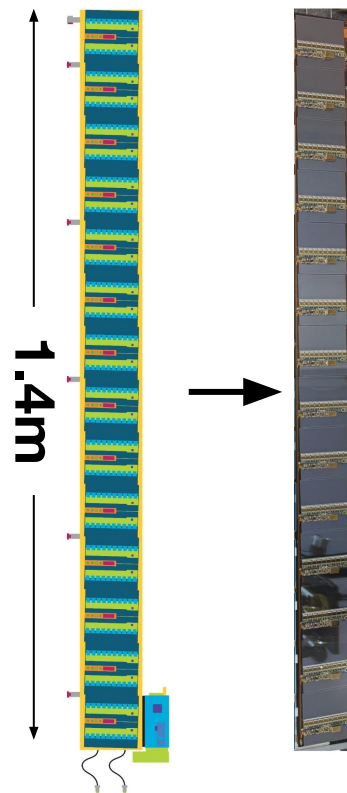


Side view



# Staves and Quality Control

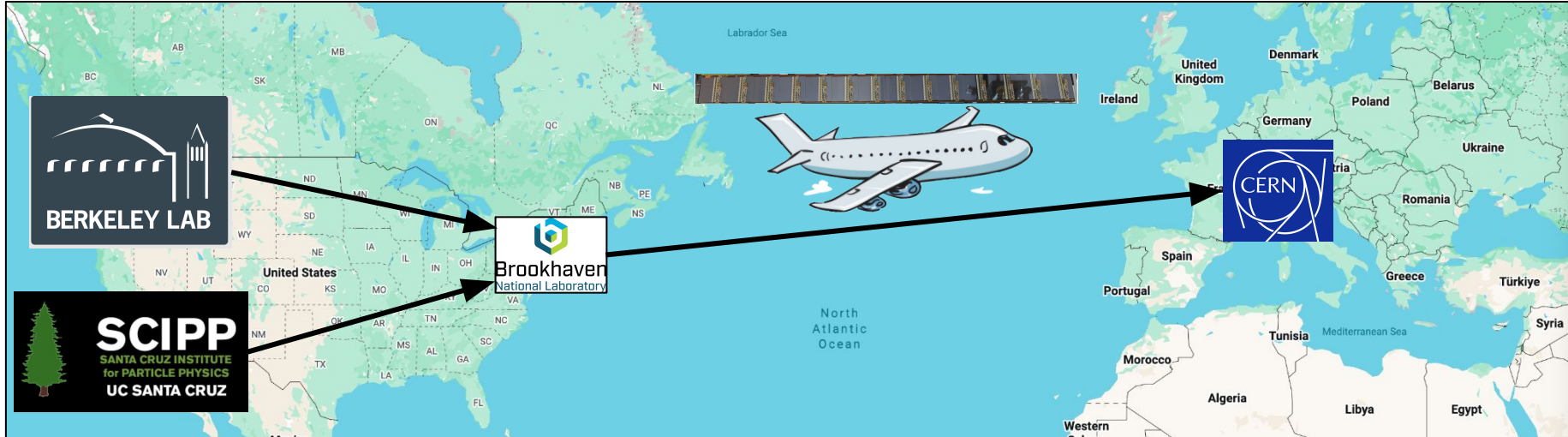
- Modules attached to **local support structures** - finished product called a **stave**
- BNL will assemble and test **196** staves to complete **half** of the Barrel (~**5500** modules)
- **Every module** and **stave** undergoes **stringent quality control**
  - Ensure a quality detector



Each stave has **14 modules per side**



# Path of a module



Produce ~ 1800  
modules each!

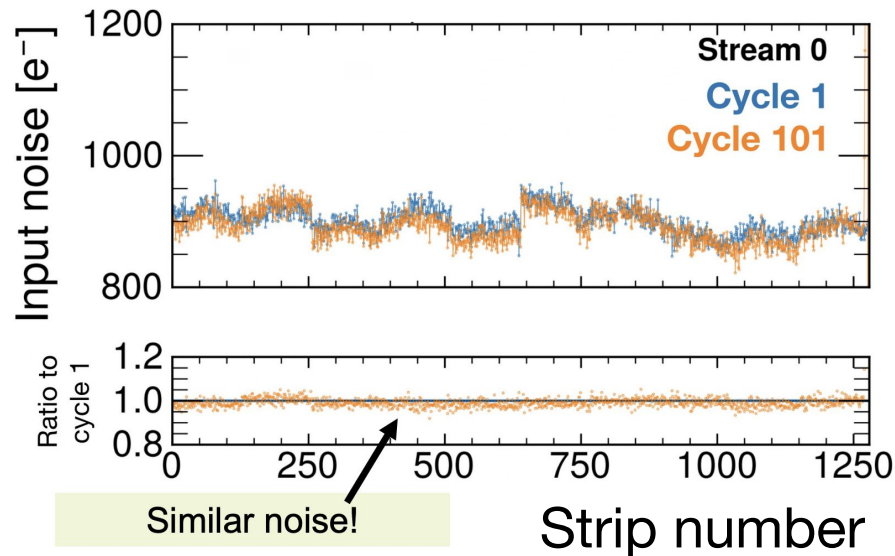
Produce ~ 1800 modules,  
load 196 staves

Send to CERN for full ITk  
integration

# Thermal cycling

- Final stage of Quality Control:  
**Thermal cycling** modules 10 times:
  - Emulate **detector** conditions (end of year shutdowns)
  - Must pass before **going on a stave**
  - I developed software to automate process. Saves **~4 person-years of manual work in the US!**
- **Extreme version** of this test - 100 thermal cycles.
  - Modules remained **operational**
  - At the time, furthered our confidence in module design...

A. Tishelman-Charny *et al* 2024 [JINST 19 P10002](#)

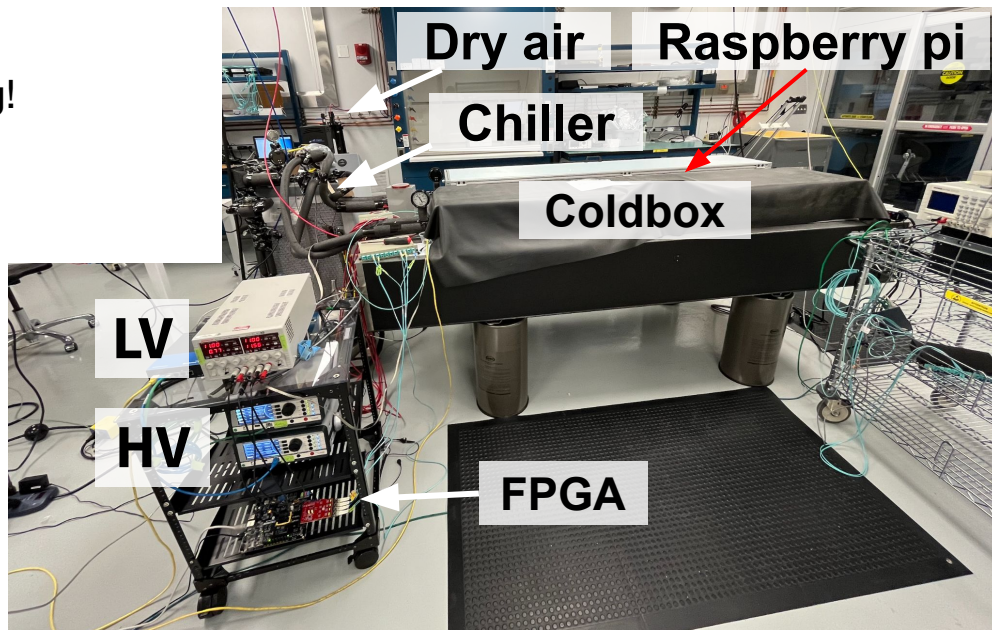


# Stave testing

- After passing, **load** module onto **stave**
- Need to check **unaffected** from **loading!**

## BNL stave testing setup

- Coldbox (Stave inside)
- Chiller (temp control)
- Dry air (humidity control)
- Power supplies to power electronics, bias sensors
- FPGA for DAQ
- Raspberry pi: Environment monitoring



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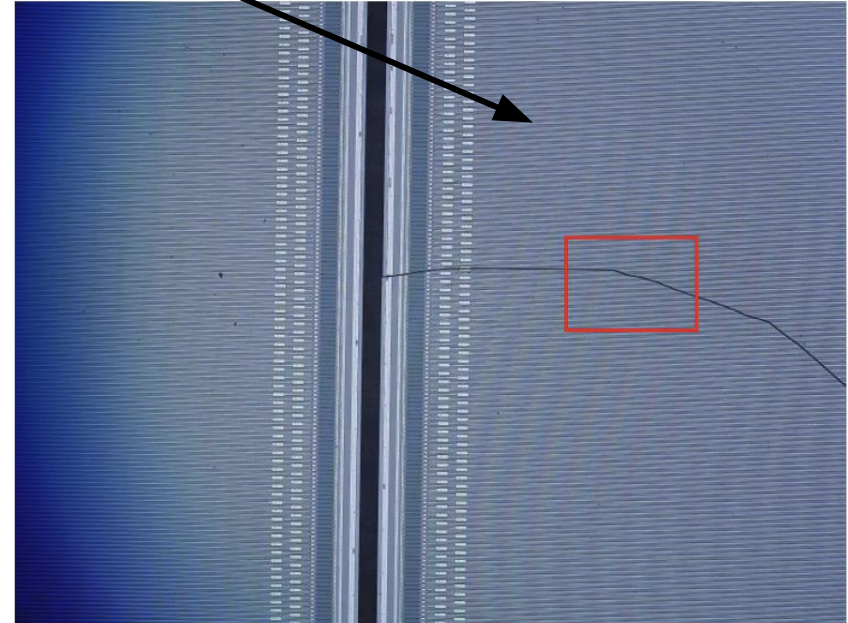
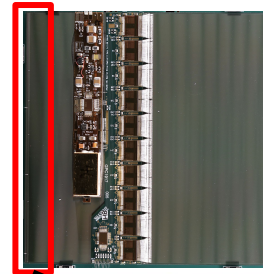
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# Fracturing

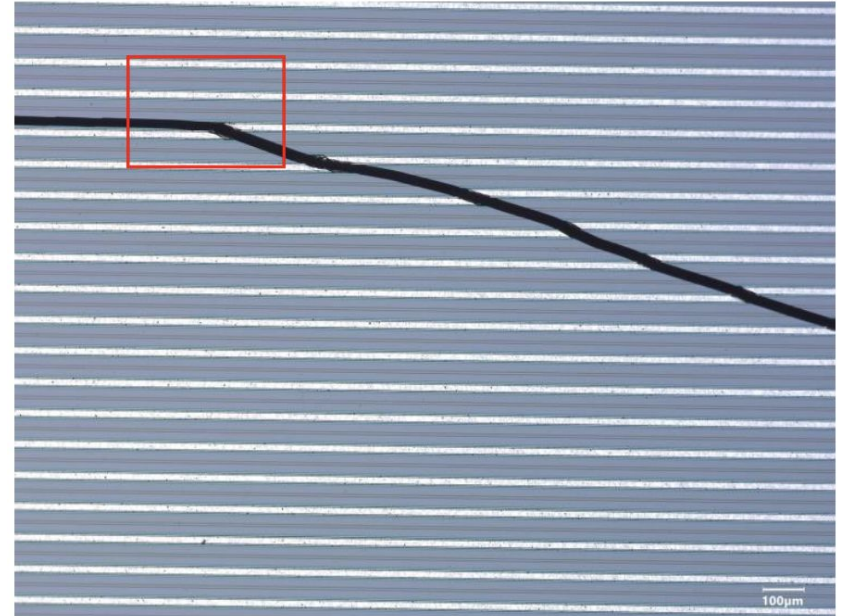
- **Pre-production:** Modules behaved well before going onto staves
- **After putting modules onto staves, and thermal cycling,** started observing electrical breakdowns
- Upon investigation, found sensors **physically fracturing**





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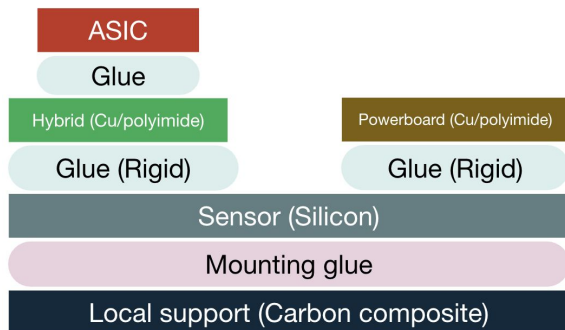
# Fracturing

- **Pre-production:** Modules behaved well before going onto staves
- **After putting modules onto staves, and thermal cycling,** started observing electrical breakdowns
- Upon investigation, found sensors **physically fracturing**
- Due to **Coefficient of Thermal Expansion mismatch**, close proximity of electronics.
- **Halted module production** while we investigated mitigation strategies

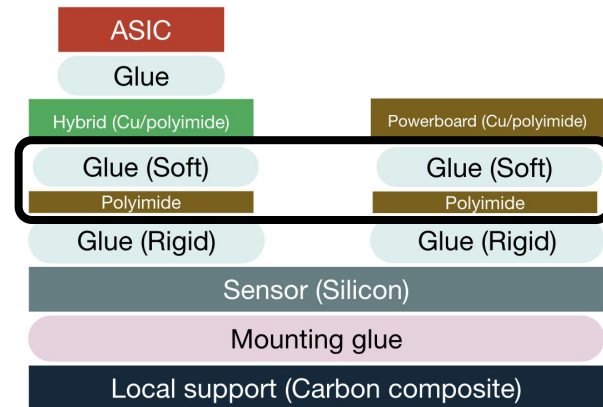


# Interposers

- Multiple module **re-designs** tested.



New  
layers



Original **mounted module** design

**Interposer** module design

- Interposer** design: Includes additional layer of **soft glue** to **absorb impact** from high stress
- Rapidly ramped up interposer **module assembly program** to fill staves and test



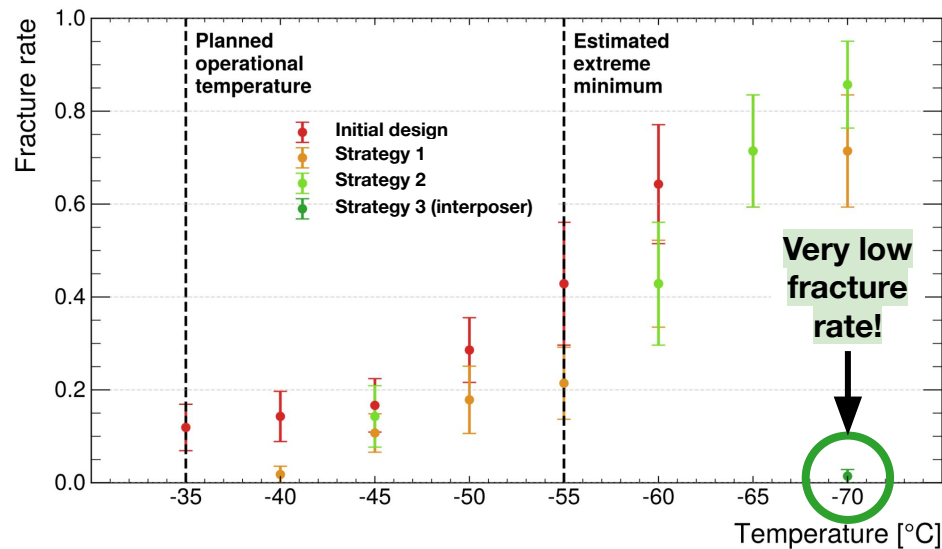
# Extended-QC

- Extended-QC performed on mitigation objects
- Extreme testing performed down to  $-70^{\circ}\text{C}$
- Best performing strategy: **Interposers** - only **1/70** modules exhibited fracturing at temperatures below extreme operations

- Moving forward with this solution, what **physics results** will be enabled at the HL-LHC?
  - Expect **HH discovery** ( $> 5\sigma$ !)
  - ~ 29%** precision on **self-coupling**
  - Improved **SH** sensitivity

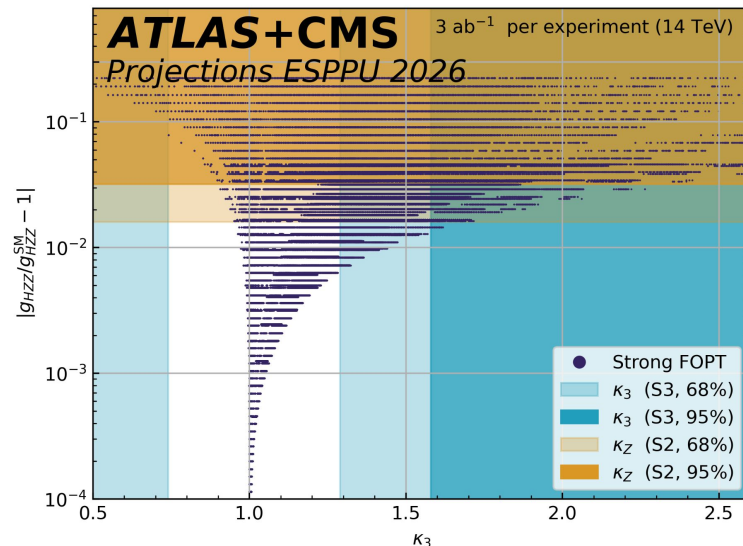


[2025 JINST 20 P09025](#)



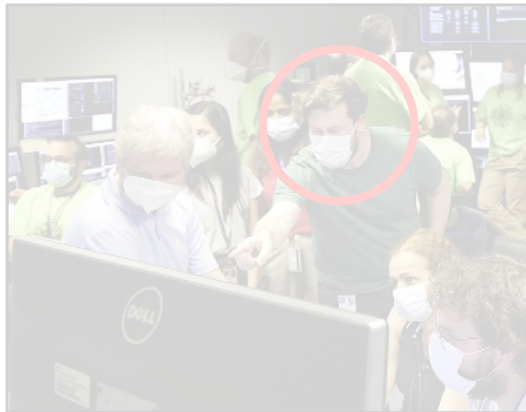
# HL-LHC projections: ATLAS+CMS

- Expect further exclusion of **BSM models**
- **Blue points:** Phase space where strong first-order phase transition is possible
- Precise Higgs coupling measurements, including the self-coupling and coupling to ZZ, **will exclude BSM scenarios**
- **Complementary to direct SH searches!**



arxiv:[2504.00672](https://arxiv.org/abs/2504.00672)

# Next section



LHC

I

7-13.6 TeV,  $\sim 500 \text{ fb}^{-1}$

**Higgs discovery**

2009-2026



HL-LHC

II

14 TeV,  $\sim 3500 \text{ fb}^{-1}$

**di-Higgs discovery**

2030-2041



Future

III

Up to  $\sim 100 \text{ TeV}$

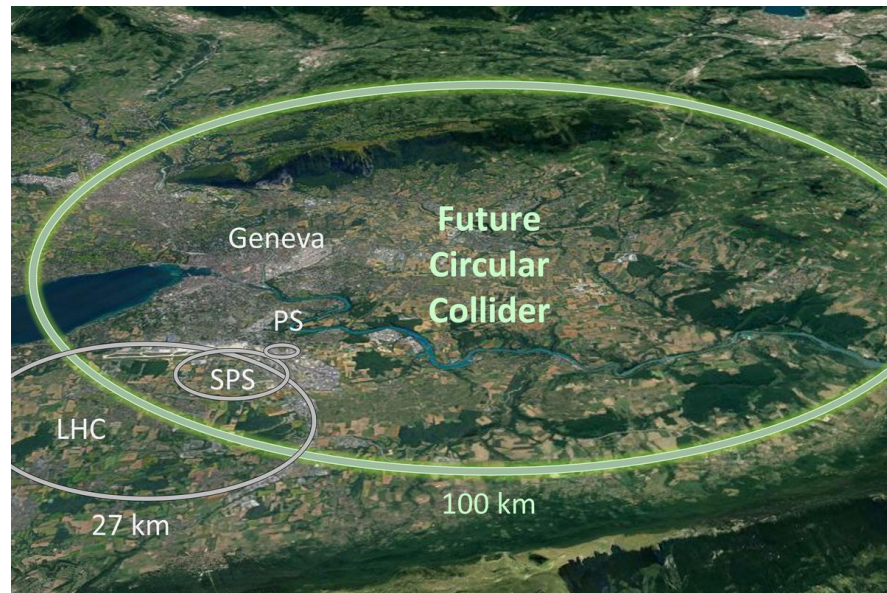
**Future discoveries**

2041 and beyond

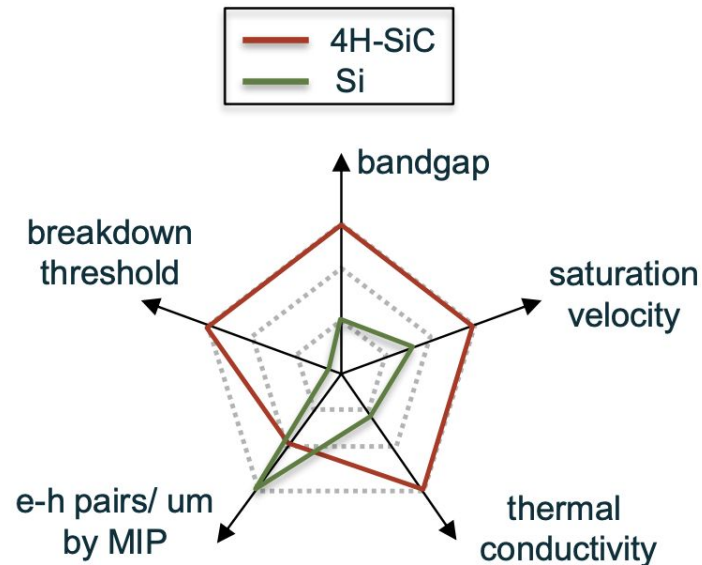
# Future colliders

- How will we push the boundaries of Higgs sector searches after HL-LHC?
- Several future collider options under consideration, e.g.:
  - CLIC, FCC-ee/hh, ILC, Muon collider, ...
  - All with different **pros and cons**
- All future detectors come with **new challenges**
  - **Need detector technology to advance alongside.**

- FCC (Future Circular Collider - ESG recommendation) would run in two stages:
  - **FCC-ee:** Lepton collider for extreme precision measurements
  - **FCC-hh:** Hadron collider for c.o.m. energy increase
- **Detectors** require cutting edge technology to meet specifications:
  - **FCC-ee:** Cleaner environment, want to **minimize interaction with dead material.**
  - **FCC-hh:**
    - Up to ~ **1000** pileup expected!
    - **Unprecedented radiation hardness needed**



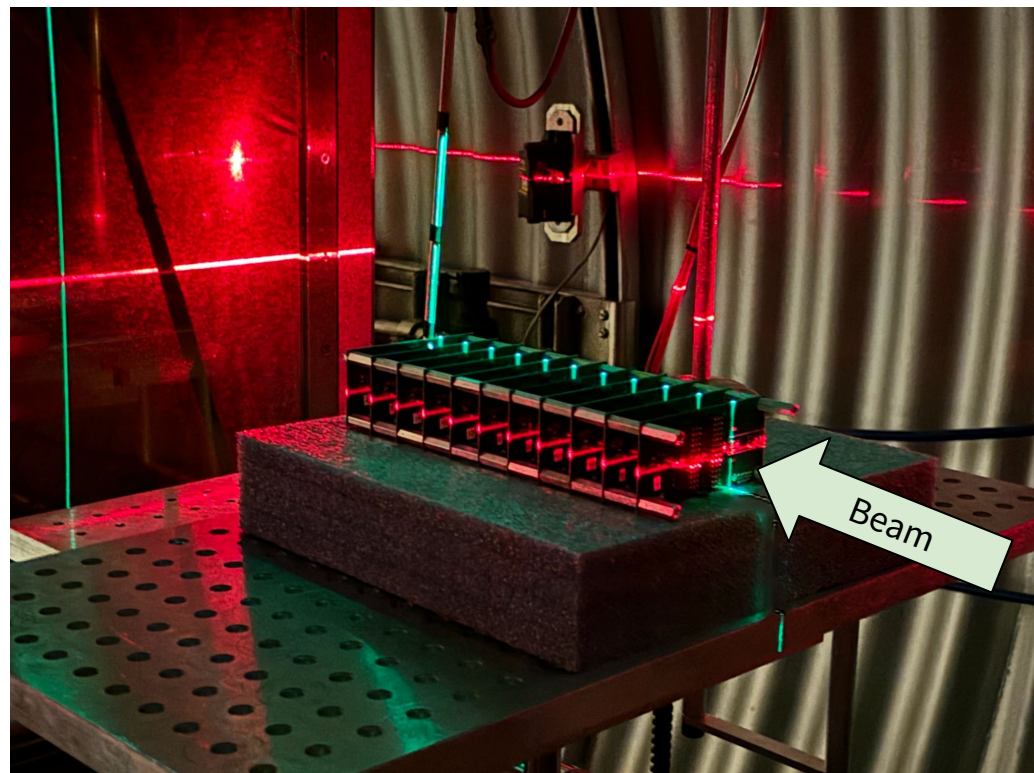
- Is there a way to **improve tracker technology** to meet these challenges?
- **Silicon Carbide (SiC):**
  - **Wide band-gap** material → Less noise
  - Expect improved **timing resolution**
  - More **radiation hard**
  - More **thermally conductive**
- Lower signal yield compared to Si, so include a **gain layer** in design
- Can operate at **room temperature:**
  - **No need for cooling system!**
  - **Huge practical benefits.** Sensor fracturing delayed ITk strips production by **about 1.5 years**





# SiC irradiation campaign

- BNL: SiC R&D in collaboration with **LBNL, NCSU**
- Irradiated **SiC LGADs** at **BNL NASA Space Radiation Laboratory (NSRL)** with **2.5 GeV protons**
- Precise alignment of beam with laser system
- Irradiated devices with different doses, including  $10^{13}$  p/cm<sup>2</sup>





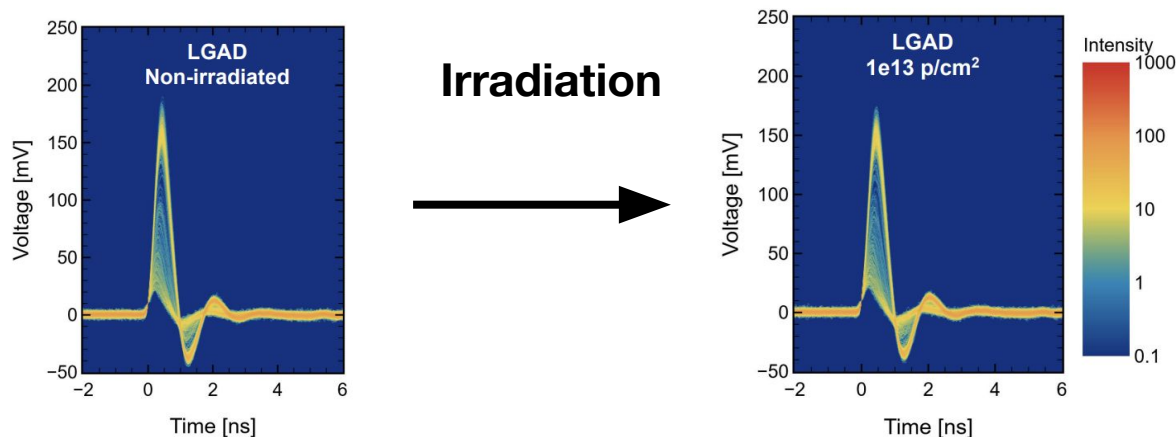
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**Carefully** remove boards after each dose



# SiC irradiation results

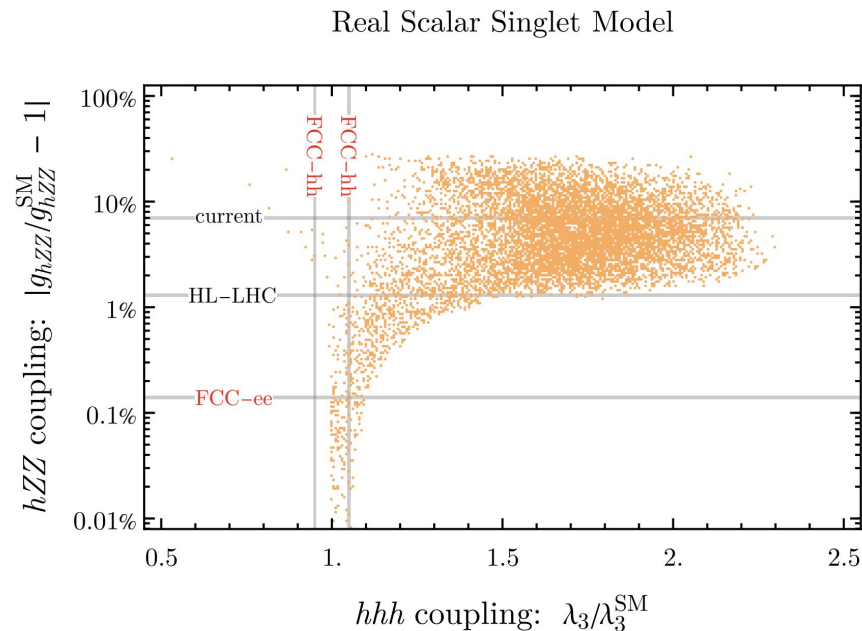


arXiv:[2507.23062](https://arxiv.org/abs/2507.23062)

- At **expected FCC-ee fluence** ( $\sim 10^{13} n_{eq}/cm^2$ ), see **very similar signal**
- Additionally, working towards ensuring **safe operation**
- Currently **further optimizing device design**

# Future results

- How will SiC help improve future **Higgs sector searches**?
  - Precision machines: **Absence of cooling system reduces material budget**
  - High energy machines: Working towards a **radiation hard system**
- **FCC-ee:**
  - Up to **per-mil Higgs coupling measurements**. Can compare to SM, indirectly rule out BSM.
  - When **combined** with HL-LHC, expect self-coupling precision of **~18%**
- **FCC-hh:** Expect self-coupling precision of **~2-3%!**



[arXiv:2505.00272](https://arxiv.org/abs/2505.00272)



# Summary



# Summary

- SM leaves **fundamental questions unanswered**
  - Use the **Higgs as a tool** to answer these questions
  - **Tracking performance crucial** to achieve this.
- **Exploring Run 3 phase space:** Further ruling out BSM models and working towards HH discovery
- Working towards completing the ITk for optimal b-tagging, HH discovery, **~29% self-coupling sensitivity**
- **Will continue at a future collider:**
  - Silicon carbide is a **promising tracker technology** - can help achieve precise FCC-ee measurements: Combine with HL-LHC for **~18%** self-coupling precision
  - FCC-hh: **~2-3% self-coupling** precision, deeper BSM phase space search

# Future plans

- **LHC:**

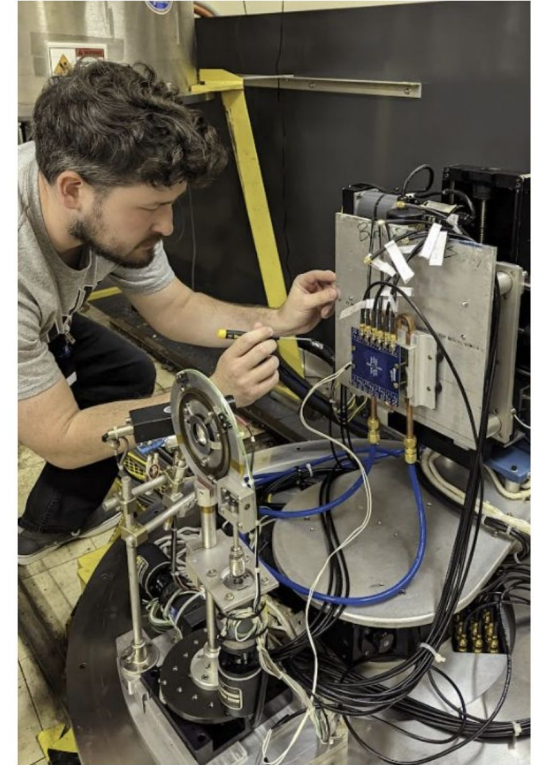
- Exploit **full Run 3 dataset** (2022-2026) to maximize **(S/H)H $\rightarrow$ bb $\gamma\gamma$**  sensitivity
- **Combination** of SH/HH - take **interference** into account.

- **HL-LHC:**

- Complete and operate the ITk to **enable di-Higgs discovery** and **upgraded SH searches**
- Finish production through efficient **testing**, solving problems during **integration** and **installation**
- Lead **commissioning of the ITk** based on previous commissioning experience

- **Future:**

- Push boundaries of **tracker technology** to further **understand the universe through the Higgs boson**
- Big plans for future colliders, but need to **ensure we can take quality data**
- Continue **Si/SiC** tracker R&D for future colliders



The background is a dark blue gradient. On the left, there is a vertical bar with a light blue top section and a yellow bottom section. The main area contains faint, semi-transparent geometric shapes, including a large cone pointing towards the center, several rectangular prisms, and a network of thin lines radiating from a central point. The text "Thank you for your attention!" is centered in a bold, white, sans-serif font.

**Thank you for your attention!**